Logical Inferences and Grammaticality

An Experimental Perspective

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Motivation

- **Problem:** Standard theories predict a strong link between NPI licensing and downward inferences.
- Experimentally, that link looks weak at best.

- **Gap:** Prior work has not tested whether the locality conditions on NPI licensing also shape inference patterns.

- **Goal:** Test the relation between NPI licensing and monotonicity inferences in double-licensing environments.
- This lets us compare licensing theories and speak to modularity.

Roadmap

1. Background (modularity, NPIs, monotonicity)
2. Method (experiment: predictions, design, ...)
3. Results
4. Interpretation
5. Conclusion

Background

- modularity of grammar and its roots
- structuralists distinguished levels such as phonology, morphology, syntax, and semantics
    - each level is analyzed autonomously

- Chomsky inherited and reframed this view (Chomsky (1965), Chomsky (1981), among others).
- **Autonomy of syntax:** syntax is independent of semantics and phonology.
- Canonical example:

(1) Colorless green ideas sleep furiously.

- Related philosophical version: Fodor (Fodor (1983)).

Why modularity?

- It explains “negative universals” such as:

1. Phonology sensitive to syntactic features (not attested)

- e.g. stress assigned directly by syntactic category

2. Syntax sensitive to phonological properties (not attested)

- e.g. word order determined by syllable count

3. Semantics sensitive to phonological form (not attested)

- e.g. meaning determined directly by vowel quality

Modularity and Negative Polarity Items (NPIs)

- NPIs occur across languages.
- They are licensed under negation and certain other operators, such as questions and conditionals.
- Examples: English *any*, Czech *sebemenší* ‘slightest’.

- Every spoken language that semanticians have examined contains Negative Polarity Items (NPIs), such as English *any* or Czech *sebemenší* ‘slightest’, which are licensed only in the scope of negation and certain other operators, such as questions and conditionals:

(2) a. I don’t have any idea how to solve this problem.
b. #I have any idea how to solve this problem.

(3) a. Do you have any idea how to solve this problem?

b. If you have any idea how to solve this problem, please let me know.

c. Every student who has any idea how to solve this problem, please raise your hand.

vs.

(4) a. #I have any idea how to solve this problem.

b. #If you have some idea how to solve this problem, please call anyone.

c. #Some student who has any idea how to solve this problem, please raise your hand.

- After some debate about whether NPIs are licensed by syntax or semantics, the consensus has been that they are licensed semantically (e.g., Ladusaw (1979), Chierchia (2013), Crnič (2011), Gajewski (2011), Barker (2018), Heim (1984), Von Fintel (1999), ...).
- The standard textbook approach links NPI licensing to downward-entailing (DE) environments (Ladusaw (1979); Von Fintel (1999); ...).

- upward and downward monotonicity came to linguistics from logic and mathematics (calculus: numerical orderings, order theory: set-inclusion):

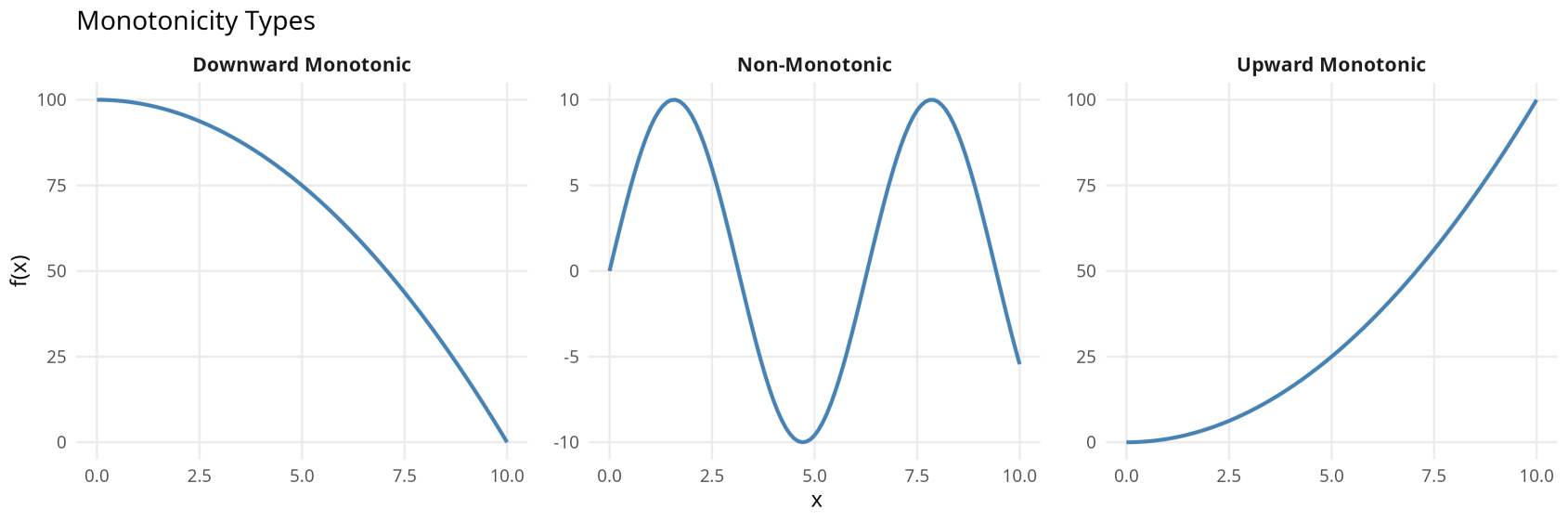


Figure 1: Monotonicity

- In formal semantics, monotonicity has been studied extensively since Barwise and Cooper (1981), but it was systematically applied to NPIs by Ladusaw (1979).

- Despite many formalizations, the overall idea is that NPIs are licensed in downward-entailing environments, that is, environments in which the following holds (from Barwise and Cooper (1981) for determiners; general definition also in Von Fintel (1999), among others):

(5) If $D(A, B)$ is true and $C \subseteq A$, then $D(C, B)$ is true – for the first argument of the determiner.

- or generally in environments where the usual $(p \wedge q) \rightarrow (p \vee q)$ (or in set terms: $A \cap B \rightarrow A \cup B$) is reversed

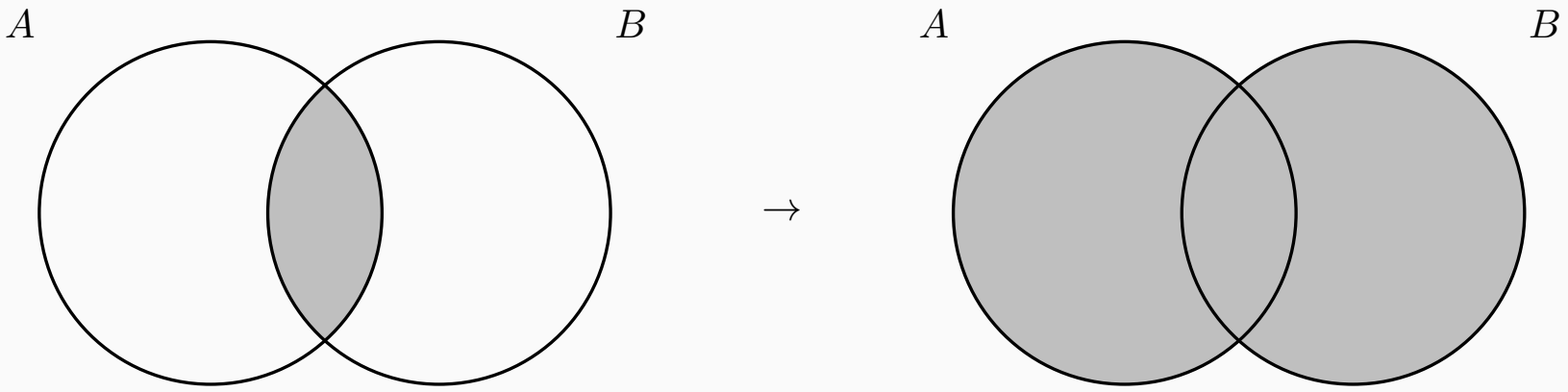


Figure 2: $A \cap B \rightarrow A \cup B$

(6) John bought red wine.

a. $\rightarrow$ John bought wine.

b. $\rightarrow$ John bought red (object).

- simple episodic sentences are upward entailing: from subsets (intersections) to supersets (unions)

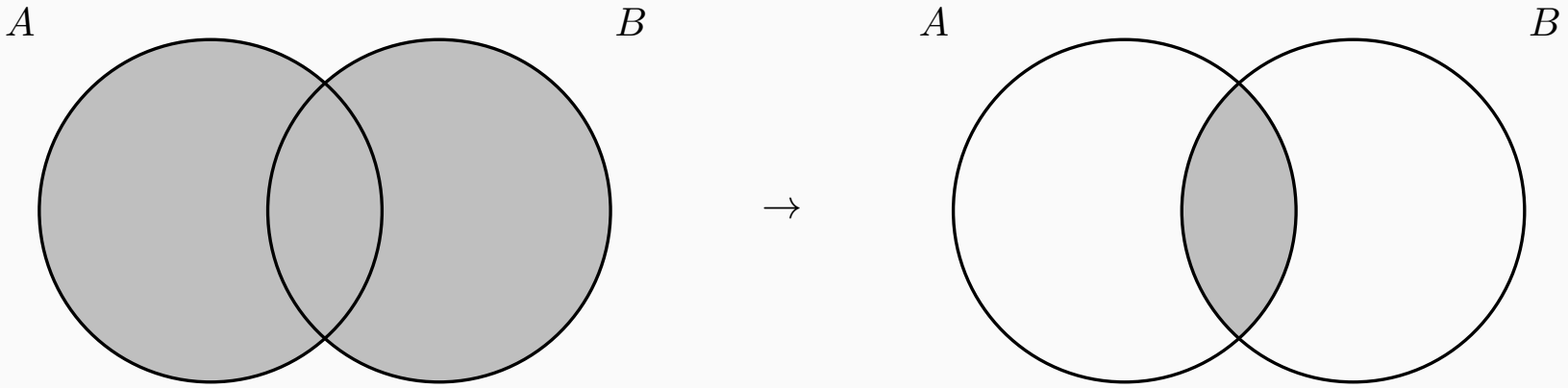


Figure 3: $A \cup B \rightarrow A \cap B$

- Negation, however, reverses monotonicity (as in logic: $\neg(p \vee q) \rightarrow \neg(p \wedge q)$):

(7) John didn't buy wine.

a. $\rightarrow$ John didn't buy red wine.

- *every* unlike *some* is downward entailing in its first argument, therefore it licenses NPIs:

(8) Every student slept.

a. $\longrightarrow$ Every black-haired student slept.

(9) Some black-haired student slept.

a. $\longrightarrow$ Some student slept.

(10) a. Every student who has any idea how to solve this problem, please raise your hand.

b. #Some student who has any idea how to solve this problem, please raise your hand.

Back to modularity

- NPIs are non-modular at least in the strict sense of modularity (syntax vs. semantics) since their licensing is determined by the semantics and entailment properties of the environment, not by syntactic structure alone
- therefore it is like syntax is directly sensitive to semantics (and our logical reasoning), which is a problem for modularity (in its strict sense)
- empirical question: do NPIs really correlate with downward inferences in the way the standard theory predicts?

Szabolcsi et al. (2008): The Null Result

- **Experiment:** Verified inferences in DE and UE environments with/without NPIs.
- **Expectation:** NPIs should facilitate the verification of DE inferences, that is, speed up reasoning in DE environments.

Table 1
Examples of the stimuli used in Experiment 1

	***Ever* example**	***Any* example**
Context sentence	Our camp is on Staten Island.	After winning a game at a local fair, children get to choose a prize.
(a) decreasing with NPI	Almost no campers have ever had a sunburn or caught a cold.	Almost no child chose any mint chocolates or decided on the cotton candy.
(b) decreasing without NPI	Almost no campers have had a sunburn or caught a cold.	Almost no child chose mint chocolates or decided on the cotton candy.
(c) non-decreasing with NPI	Almost every camper has ever had a sunburn or caught a cold.	Almost every child chose any mint chocolates or decided on the cotton candy.
(d) non-decreasing without NPI	Almost every camper has had a sunburn or caught a cold.	Almost every child chose mint chocolates or decided on the cotton candy.
Inference question	Would it be reasonable to say that almost no [almost every] camper[s] have [has] caught a cold?	Would it be reasonable to say that almost no [almost every] child decided on the cotton candy?

Figure 4: Szabolcsi et al. (2008) results

- **Result: No evidence found.**
  - "Expectations did not match results and we were unable to find evidence for the recognition." Szabolcsi, Bott, and McElree (2008)
- **Implication:** NPI licensing and monotonicity inference might not be directly linked in processing.

Given the existing evidence on NPI licensing, it seems almost like a foregone conclusion that the processor should recognize the relationship between monotonicity and NPI licensing. Indeed, when we started this project, our intention was to investigate how this recognition occurred, rather than whether it occurred. As is often the case, however, expectations did not match results and we were unable to find evidence for the recognition.

Chemla et al. (2011): The Correlation

- **Method:** Correlated individual monotonicity judgments with NPI acceptability.
  - Used French NPI *le moindre* (similar to Czech *sebemenší*).
  - Varied quantifiers (no, some, every, exactly 12...).
- **Result: Positive Correlation.**
- The inferences a speaker considers valid predict NPI acceptability Chemla, Homer, and Rothschild (2011).
- *Note:* Chemla et al. (2011) found Kendall's $\tau \approx 0.28 (p = .032)$ between subjective monotonicity judgments and NPI acceptability. This corresponds to a Bayes Factor of approximately 3-6, indicating substantial, though not strong, evidence for a correlation.
- **Significance:** First positive evidence linking semantic intuition to NPI licensing (but not very strong).

Denić et al. (2021): Recent Evidence

- **Design**: Tested PI effects on monotonicity judgments in NM and doubly-negative (DN) environments.

(11) a. No alien spent a year without seeing (some/any) doves.
b. No alien spent a year without seeing birds.

(12) a. Exactly 12 aliens saw any doves.
b. Exactly 12 aliens saw any birds.

- **Key Findings** of Denić et al. (2021):
  - In **NM environments**, NPIs reduced perceived upward-entailingness by ~4-5 percentage points (directional ratings: ~55% vs. ~60% baseline); PPIs had no effect.
  - In **DN environments** (globally UE but locally DE), effects were mixed: meta-analysis suggests small NPI/PPI influences, but individual experiments showed variability.
  - **No effects** in clear DE/UE environments (likely ceiling effects).
- **Conclusion**: Polarity items *do* influence inferential judgments, but only in ambiguous contexts and with small effect sizes—supporting a graded, subjective link between monotonicity perception and PI licensing, not a simple "flag" mechanism.

Motivation: Why Re-examine the NPI–Inference Link?

- Theoretical tension:
  - Classic view (Ladusaw 1979): NPIs are licensed in Downward Entailing (DE) contexts → hypothesized to act as *processing cues* for downward inferences.
  - Experimental reality: Mixed evidence. Szabolcsi et al. (2008) found **no effect**; Chemla et al. (2011) and Denić et al. (2021) found **small, subjective correlations** ($\tau \approx 0.28$; ~4–5% scale shifts).
- Open question:
  - If NPIs signal monotonicity, *where* and *how* should this signal operate?
  - Prior work tested global monotonicity, but **not the role of structural locality** in licensing vs. inference.

- **Our focus**:
- Double Licensing (DL) environments (*Petr doubts that… not… NPI*) allow us to isolate whether NPI licensing and inferential reasoning are sensitive to the **same structural configurations**.
- We compare licensing and inference tasks with matched structures and test for a correlation.

Motivation: The Locality Gap

- **Prior Work (Denić et al. 2021):** Found small NPI-inference correlations ($\tau \approx 0.28$) but tested **global** monotonicity.
- **Open Question:** Do licensing and inference rely on the *same structural locality*?
- **Our Focus:** Double Licensing (DL) environments (*doubt... not... NPI*) isolate polarity domains (Homer 2021).

Double Licensing (DL) Environments

- Examples discussed by Denić et al. (2021) include cases such as *Every alien who did not see any doves is hairy*:

(13) No alien spent a year without seeing any doves.

- These cases involve a single polarity domain in the sense of Homer (2021), and although NPIs can be licensed at a distance, their licensing is still subject to locality requirements:

(14) I don't think [that he believed [that she mentioned [that they found any errors]]].

Homer (2021) studies locality constraints, and one of his key findings is:

- an NPI must be c-commanded by a DE operator within its local polarity domain
- when two DE operators are in the same domain, they do not compose to license an NPI in a different domain
- which explains the following contrast:

(15) a. #I don't doubt that he bought any books.
b. I doubt that he didn't buy any books.

- Following Homer (2021), the experiment directly tests the locality manipulations that determine NPI licensing and asks whether they affect inference reasoning in the same way, and whether the two are correlated.

Contribution: Dissociation & Scope

- **Design:** Parallel tasks (licensing + inference) with matched DL structures (DOUBT-NEG vs. NEG-DOUBT).
- **Result: Strong Dissociation.** Licensing is domain-sensitive (BF > 10^{15}); Inference is not (BF ≈ 0.04).
- **Implication:** No speaker-level correlation. Supports **Barker (2018)**: NPIs mark narrow scope, not global monotonicity.

Research Question

- **RQ1:**

(16) Do NPIs in double licensing environments correlate with facilitating downward inferences?

- **Hypothesis:**

a) under the standard DE theory, we expect a correlation between NPI licensing and downward inferences in DL environments;

b) under Barker's (2018) alternative scope theory of NPI licensing, we expect no correlation between NPI licensing and downward inferences in DL environments.

Data

- Source: original experiment
- joint work with Lucie Chumchalová (now at the University of Potsdam)

Figure 5: Lucie Chumchalová

2 experiments (both acceptability judgment tasks):

- 7-point Likert scale
- Online on *L-Rex* platform

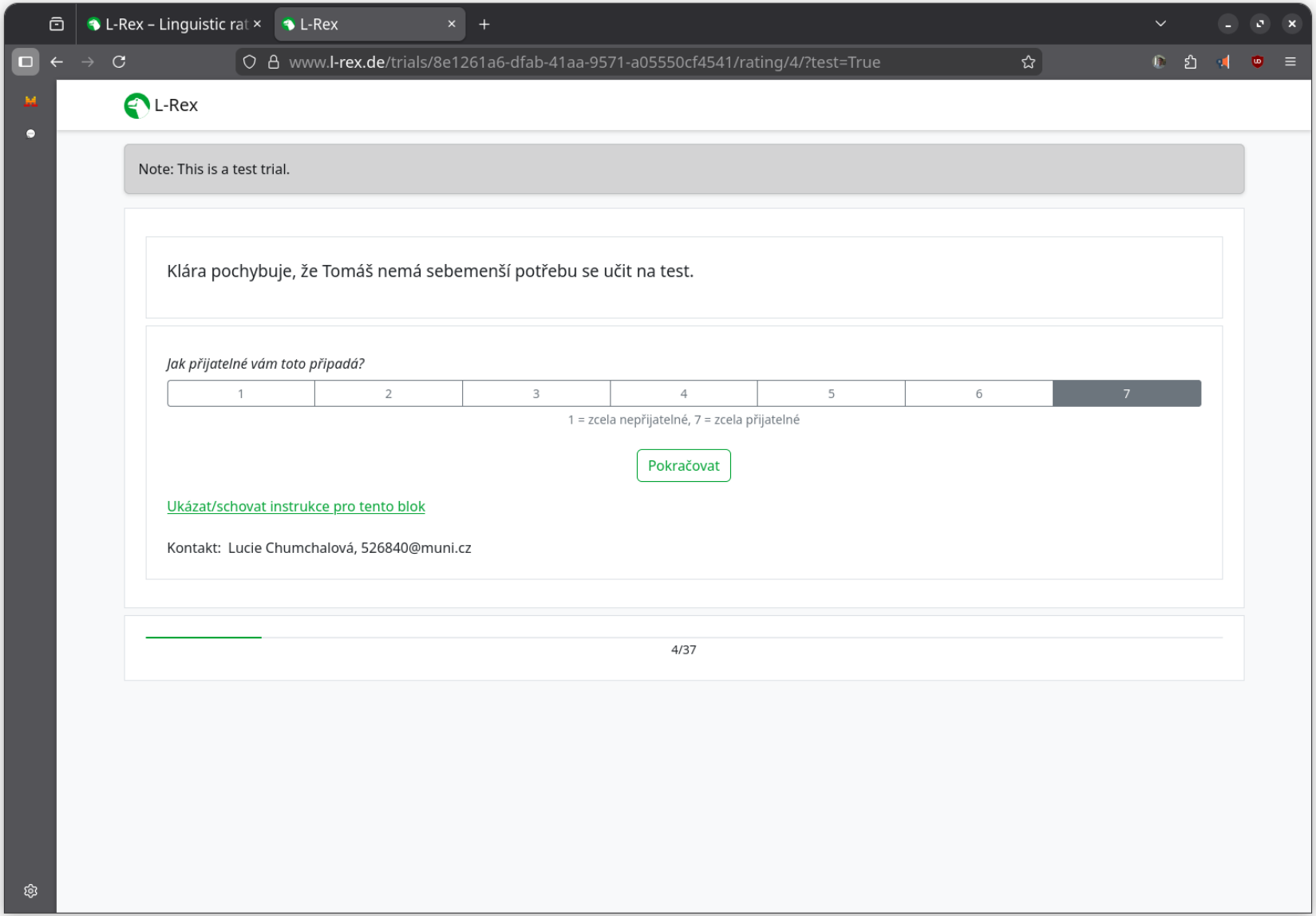


Figure 6: Example trial

- 57 of the 70 native Czech speakers passed the filler items and were included in the analysis
- Two versions with reversed order to control for order effects

Experimental Items

BASELINE:

(17) Policie **ne**-má *sebemenší důvod* tu stopu
Police neg-have.3sg slightest reason that.acc trail.acc
zahladit.
cover.
'The police have no reason to cover the trail.'

DOUBT-NEG:

(18) Petr **pochybuje**, [že policie **ne**-má *sebemenší*
Petr doubt.3sg that police neg-have.3sg slightest
důvod tu stopu zahladit].
reason that.acc trail.acc cover.
'Petr doubts that the police have no reason to cover the trail.'

NEG-DOUBT:

(19) Petr **ne-pochybuje**, [že policie má *sebemenší*
Petr neg-doubt.3sg that police have.3sg slightest
důvod tu stopu zahladit].
reason that.acc trail.acc cover.
'Petr does not doubt that the police have the slightest reason to cover the trail.'

Experiment 1: NPI Licensing in DL Environments

1×3 design:

- BASELINE
- DOUBT-NEG
- NEG-DOUBT

Experiment 2 Examples

DE conditions:

(20) Petr **pochybuje**, [že si Marie **ne**-pořídila kočku].
'Peter doubts that Marie didn't get a cat.'
Petr doubt.3sg that refl Marie neg-got.3sg cat.acc.

a. → siamese cat [DE-DOWN]

b. → animal [DE-UP]

- locally (where the object we ask about is) downward entailing
- the experimental items contained full sentences (e.g., *Peter doubts that Marie didn't get a Siamese cat*)

UE conditions:

(21) Petr **ne-pochybuje**, [že si Marie pořídila kočku].
Petr neg-doubt.3sg that refl Marie got.3sg cat.acc.
'Petr does not doubt that Marie has a cat.'

a. → siamese cat [UE-DOWN]

b. → animal [UE-UP]

- locally (where the object we ask about is) upward entailing
- locally, at the position of the object under discussion, the environment is upward entailing

Experiment 2: Inference Reasoning

2×2 design:

- de/ue (local environment)

vs.

- up/down (inference type)

=

- DE-DOWN
- DE-UP
- UE-DOWN
- UE-UP

Method

- standard descriptive statistics together with Bayesian regression modeling
- Bayesian statistics are useful here because they allow us to assess evidence for the null hypothesis directly, unlike the frequentist approach, where we can only estimate the probability of the data given the null hypothesis (p-value) and then infer support for the null only indirectly

Results: Descriptive statistics

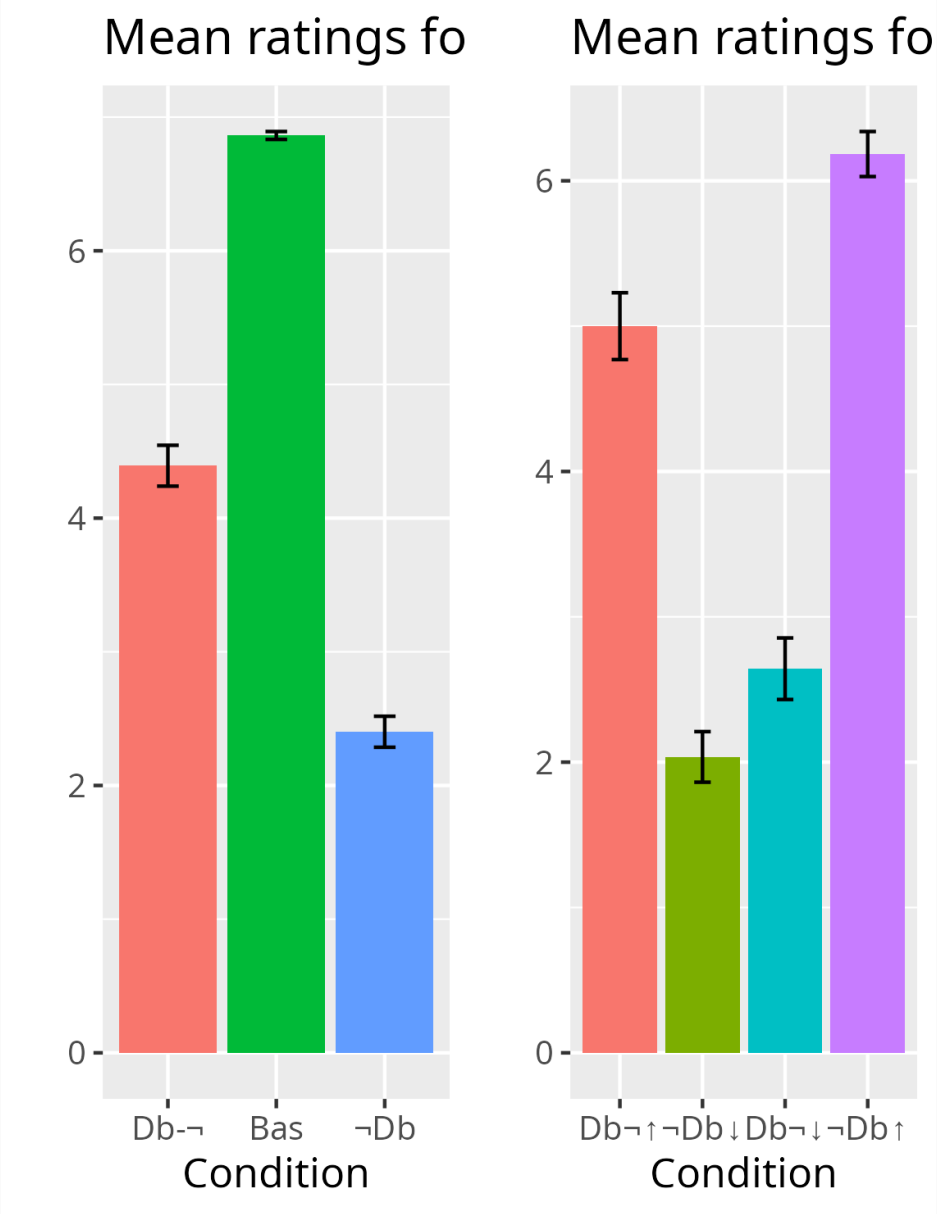


Figure 7: Acceptability and Inferences in DL environments

Results: Experiment 1 model

Bayesian linear regression models in *R* (R Core Team (2021)), package *rstanarm* (Brilleman et al. (2018))

- dependent variable: acceptability rating (1-7)
- independent variable: experimental condition (baseline, doubt-neg, neg-doubt)
- reference level: doubt-neg (Petr **pochybuje**, [že policie **ne**-má *sebemenší důvod* tu stopu zahladit])
- random intercepts for participants and items (more complex models with random slopes did not converge)

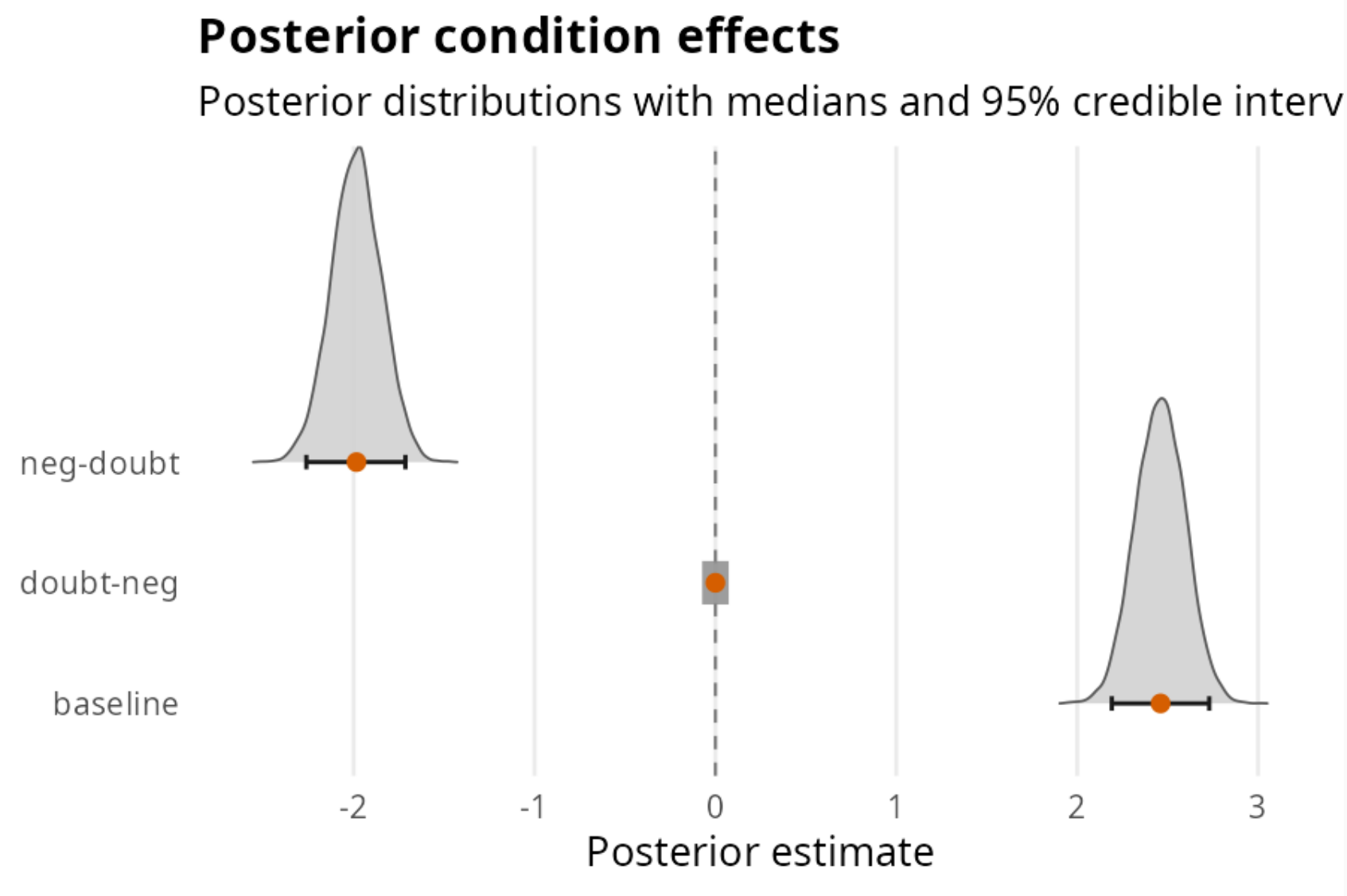


Figure 8: Posterior distribution plot

Model 1 (NPI Licensing):

- BASELINE is more natural than DOUBT-NEG
  - $\widehat{\beta} = 2.46$, BF 1.01e+18
    - extreme and decisive evidence for the alternative hypothesis
- NEG-DOUBT is less natural than DOUBT-NEG:
  - $\widehat{\beta} = -1.98$, BF 2.25e+15
    - extreme and decisive evidence for the alternative hypothesis

Model 2 (Inferences)

- the design used 2x2 coding:
  - ENV: DE vs. UE (local environment) = position of the negation (DE: Petr **pochybuje**, [že si Marie **ne**-pořídila kočku]. vs. UE: Petr **nepochybuje**, [že si Marie pořídila kočku])
  - INF: UP vs. DOWN (inference type) = whether the inference is upward (e.g., Marie got a cat → Marie got an animal) or downward (e.g., Marie got a cat → Marie got a Siamese cat)

- dependent variable: acceptability rating (1-7)
- independent variables: ENV and INF, and their interaction
- sum coding for the independent variables
- random intercepts for participants and items (more complex models with random slopes did not converge)

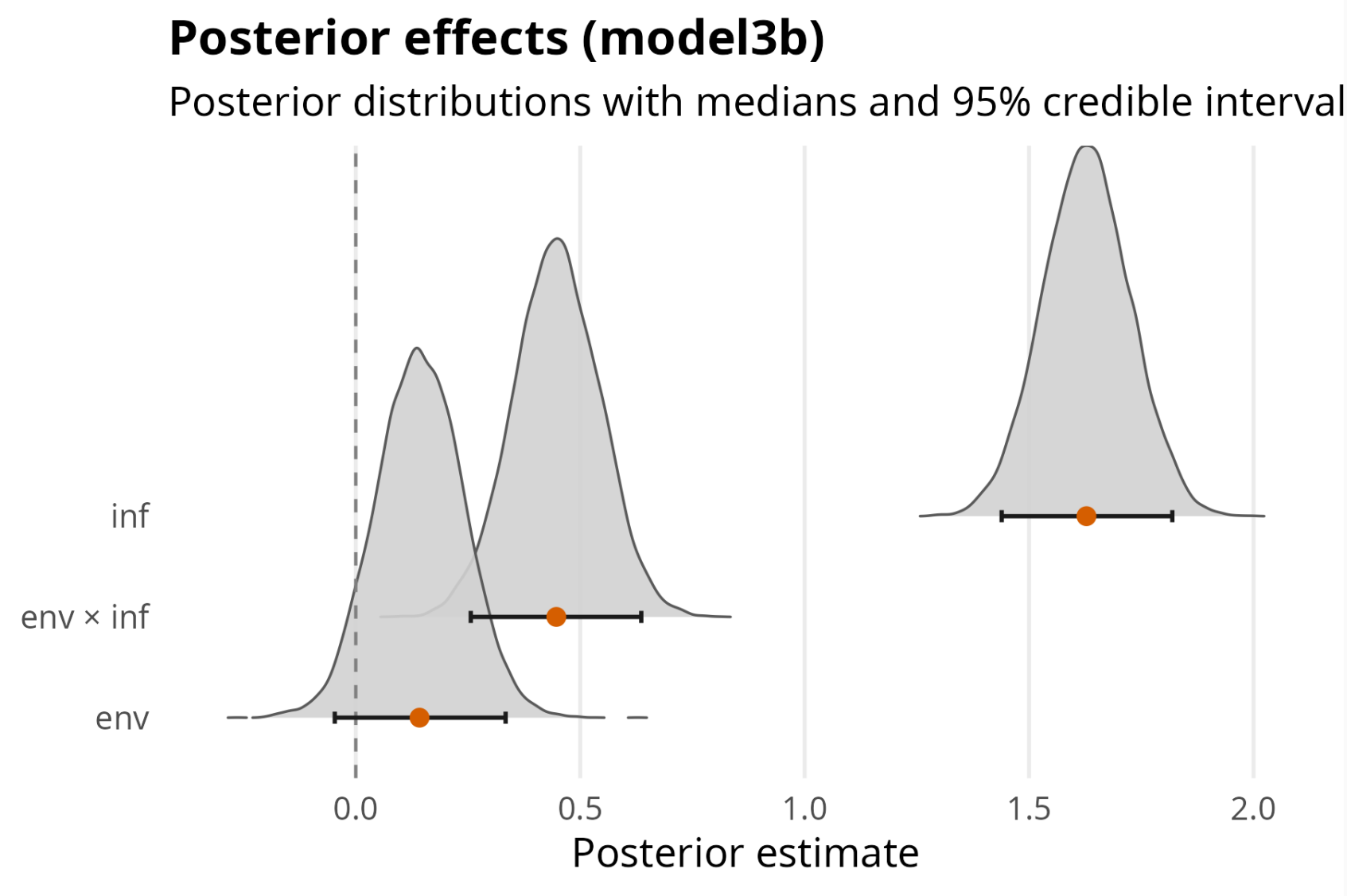


Figure 9: Posterior distribution plot (model2)

- Upward inferences strongly preferred
  - INF: $\hat{\beta} = 1.63$, BF 1.38e+16
    - extreme and decisive evidence for the alternative hypothesis
- No effect of negation position
  - ENV: $\hat{\beta} = 0.14$, BF 0.044
  - strong evidence for the null hypothesis: $1/0.044 = 22.7$, that is, the null is 22.7 times more likely than the alternative
- Intriguing interaction INF × ENV!
  - but much weaker effect: $\hat{\beta} = 0.45$, BF 146.47
    - strong evidence for the alternative hypothesis, but not as extreme as the main effect of INF

Interpretation

Key Findings

- **Answer to research question:**

(22) Do NPIs in double licensing environments correlate with facilitating downward inferences?

- NPI licensing in DL environments does not correlate with the facilitation of downward inferences

- the effect of INF was extreme and decisive: in DL environments, participants overwhelmingly preferred upward inferences
    - there was no effect of negation position on this inference pattern
    - yet the position of negation is exactly what determined NPI licensing in the first part of the experiment
        - as shown by the extreme evidence for the DOUBT-NEG vs. NEG-DOUBT contrast

The interaction between ENV and INF

- the interaction between ENV and INF is weaker ($\widehat{\beta} = 0.45$, BF 146.47) than the other effects, but it is clearly present
- it indicates that the preference for downward inferences is stronger in the DE environment (DOUBT-NEG), which is also the environment that licenses the NPI, that is, the one with negation local to the NPI clause

- this is the only part of our experimental evidence that points in the direction of the findings reported by Denić et al. (2021) and Chemla, Homer, and Rothschild (2011), but because it is weaker than the other effects, we interpret it as a possible diachronic effect in the sense of Herburger (2023):
- while NPIs may have grammaticalized as markers of downward monotonicity, they are not necessarily processed as such synchronically, and the interaction effect could be a trace of this diachronic process (another possible explanation is an NPI illusion)

The theory which works

- the scope theory of NPI licensing proposed by Barker (2018), which can account for NPI licensing in the local domain (and also in non-monotonic contexts)
- NPIs are licensed if their wide scope does not entail their narrow scope:
    - licensing by negation
    - the NPI signals the narrow scope of the indefinite with respect to negation

- (23-a) is true if, out of three students A, B, and C, there is one that John did not see, whereas (23-b) is false in that scenario

(23) John didn't see *any* students.

a. $\exists x[student'(x) \wedge \neg see'(J, x)]$

b. $\not\models \neg\exists x[student'(x) \wedge see'(J, x)]$

- intuition: the NPI is a scope marker

- in a positive sentence, the two scopes are equivalent, so the NPI is not licensed:

(24) #John saw any students.

a. $\exists x[student'(x) \wedge see'(J, x)]$

b. $\models \exists x[see'(J, x) \wedge student'(x)]]$

- intuition: the NPI is functionally vacuous here (there is no scope distinction to mark), so it is not licensed

- for NPI acceptability, negation in the local domain is crucial
- in Barker's theory, the NPI is licensed if its wide scope does not entail its narrow scope within its local domain

(25) Petr **pochybuje**, [že policie **ne**-má *sebemenší důvod* tu stopu zahladit].

a. $\exists x[reason'(x) \wedge \neg cover'(police, x) \wedge slightest'(x)]$

b. $\not\models \neg\exists x[reason'(x) \wedge cover'(police, x) \wedge slightest'(x)]$

- but because it is a marker of scope, it is an unreliable signal of downward inferences

- Barker's theory also explains the licensing in non-monotonic contexts, such as:

(26) Exactly one student passed any exam.

- since Linebarger (1987), these have been considered the hardest cases for the standard monotonicity-based theory of NPI licensing

(27) Exactly one student passed any exam.

a. $\exists y\,[\text{exam}(y) \wedge \exists! x\,(\text{student}(x) \wedge \text{passed}(x, y))] \not\models$

b. $\exists! x\,[\text{student}(x) \wedge \exists y\,(\text{exam}(y) \wedge \text{passed}(x, y))]$

- countermodel: Alice passed E_1 only; Bob passed E_2 only
- wide scope of the indefinite is TRUE: E_1 was passed by exactly one student (Alice)
- narrow scope of the indefinite is FALSE: no exam was passed by exactly one student
- for Barker, the NPI is licensed because the wide scope does not entail the narrow scope

Back to the Experiment

- Barker's theory and Homer's (2021) environment-based approach explain the dissociation between NPI licensing and monotonicity inferences in DL environments:

1. NPI licensing is locality-sensitive: local negation licenses the NPI (Homer 2021), and the local domain is also relevant for the narrow-scope computation (Barker 2018); the acceptability of the DOUBT-NEG condition is much higher than that of the NEG-DOUBT condition.

2. But inferences are not sensitive to the same locality: the position of negation does not affect the inference pattern; **upward inferences are always preferred**.

- We license locally but think globally (or at least not locally).

Limitations

- the experiment does not cover non-monotonic environments like *většina* (most), which are predicted to license NPIs by Barker's theory but not by the standard monotonicity-based theory
    - Appendix: Experiment with *většina* (most) as a non-monotonic licensor of NPIs

- we only tested inferences without NPIs, but it would be interesting to test whether the presence of NPIs in the same local domain affects inference patterns, for example by making downward inferences more likely
    - experiment in progress
- other types of NPI theories include non-veridicality (Giannakidou (1997), Giannakidou (2007)), but this approach appears to face both empirical (Alexandropoulou, Bylinina, and Nouwen (2020)) and theoretical (Gajewski (2011)) problems

Conclusion

- **Takeaway 1:** The DE theory is incorrect, and our data support Barker's (2018) alternative scope theory of NPI licensing.

- **Takeaway 2:** The implications of the dissociation for modularity are complex, but the results suggest that NPI licensing reflects a combination of syntactic and semantic factors:
    - syntactic: the locality constraints on licensing (Homer 2021) – plus c-command;
    - semantic: the scope relations that determine licensing (Barker 2018) – plus entailment comparison;
    - (pragmatic: Heim (1984), Crnič (2011), ...);
- bigger picture: NPIs link syntax, semantics, and pragmatics in a complex way, thereby challenging strict modularity.

- **Next step:** Future work should examine non-monotonic environments (e.g., *většina* ‘most’) and test the effect of NPIs on inference patterns.

Thank you for your attention!

- Contact: docekal@phil.muni.cz
- Slides / repository: https://github.com/mojmirdocekal/sandbox2026
- Thanks to many people for feedback and discussion, especially Lucie Chumchalová, Rhiana Horovská, Radim Lacina, Olav Mueller-Reichau, and Giorgio Sbardolini.

Appendix

- Joint work with Rhiana Horovská

Background: The Open Question

The Classic View (Ladusaw 1979)

- NPIs are licensed in Downward Entailing (DE) contexts
- NPIs signal downward inferences

The Problem

- Experimental evidence is mixed:
  - Szabolcsi et al. (2008): **no correlation**
  - Chemla et al. (2011), Denić et al. (2021): **small, subjective correlations**
- Double Licensing (DL) environments still involve *monotonic* operators (negation, *doubt*)

The Critical Gap

- Do NPIs require DE *synchronically*, or only narrow scope?
- Need a **genuinely non-monotonic** licensing context

Why “Většina” (Most)?

Theoretical Status

- Proportional quantifier (Barwise & Cooper 1981)
- Denotation: $[\![\textit{většina}]\!](A)(B) = 1$ iff $|A \cap B| > |A \setminus B|$
- **Neither DE nor UE** in its restrictor

Prediction

- If NPIs require DE: *většina* should **not** license NPIs
- If NPIs signal narrow scope (Barker 2018): *většina* **can** license NPIs

The Test

- NPI *sebemenší* (“least/even the slightest”) in restrictor of *většina*
- Parallel acceptability and inference tasks with the same participants

Experiment Design: Two Tasks

Task 1: Acceptability (1×3 factorial)

- Quantifier: *all* (DE) / *most* (NM) / *some* (UE)
- NPI: *sebemenší vírou* ("with the least faith")
- Scale: 1–5 Likert
- N = 53 total (24 acceptability-first, 29 inference-first)

(28) *Všichni/Většina/Někteří občané/-ů se sebemenší*
all/majority/some citizens with *sebe*-small
vírou v Boha chtěli/-a nalezenému dítěti
faith in God wanted found.DAT child.DAT
vystrojit křest.
arrange.INF christening.ACC
'All/Most/Some of the citizens with even the least faith in God wanted to arrange a christening for the found child.'

Task 2: Inference Judgment (2×3 factorial)

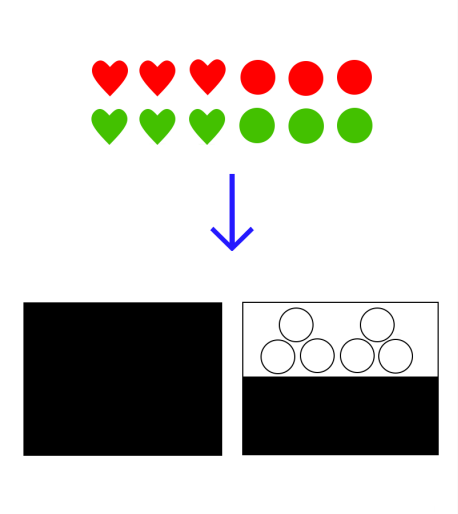
- Direction: DE (subset) / UE (superset)
- Quantifier: *all* / *most* / *some*
- Visual stimuli: shapes in boxes with concealed colors
- Response: Yes/No entailment judgment

Control

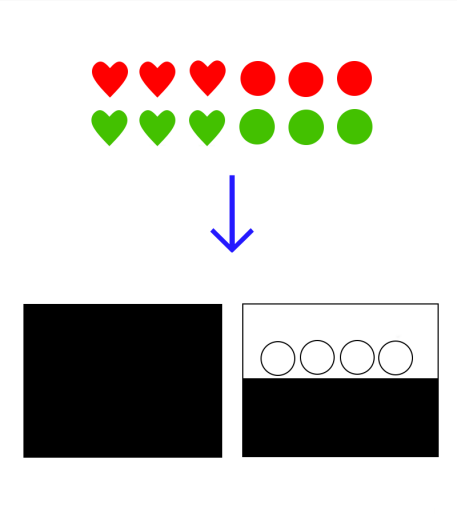
- Version order showed no effect ($BF_{01} > 4$)
- Data aggregated for analysis

(29) a. *Sentence A:* [All/Most/Some] of the circles are in the box on the right.

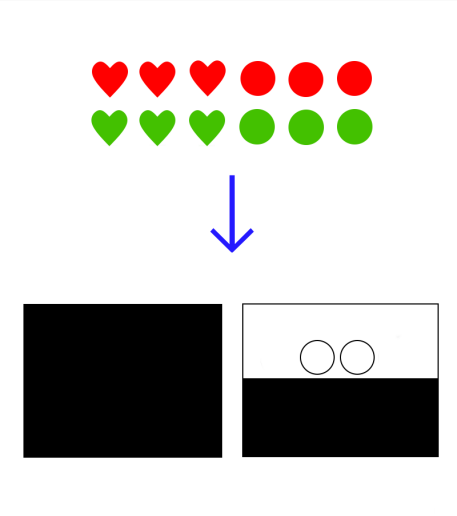
b. *Sentence B:* [All/Most/Some] of the [shapes/green circles] are in the box on the right.



(a) “all-DE” and “all-UE”



(b) “most-DE” and “most-UE”



(c) “some-DE” and “some-UE”

Figure 10: Visuals for the inference task.

Acceptability Results

Gradient Pattern

Quantifier	Mean Rating	Interpretation
All (DE)	μ = 4.34	Highest acceptability
Most (NM)	μ = 3.95	**Intermediate**
Some (UE)	μ = 3.42	Lowest acceptability

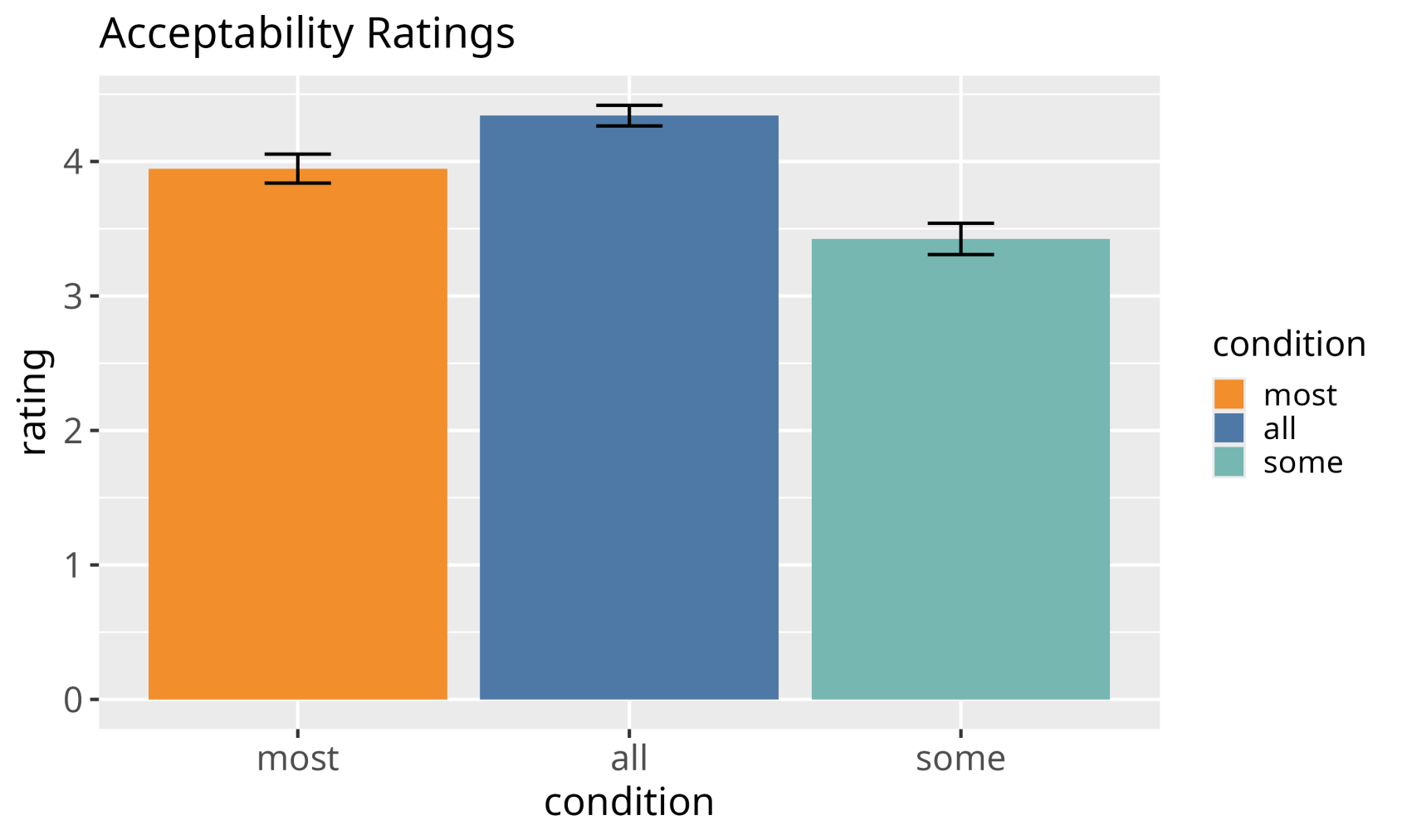


Figure 11: Acceptability ratings by quantifier

Bayesian Evidence

- *All* > *Most*: $\widehat{\beta} = 0.43$, BF_{10} = 5.14
- *Some* < *Most*: $\widehat{\beta} = -0.49$, BF_{10} = 24.6
- Both effects: pd > 99.96%

Key Finding

- *Most* licenses NPIs at **intermediate levels** despite being non-monotonic

Inference Results: Monotonicity Confirmed

DE Inferences (Subset)

Quantifier	"Yes" Rate	BF_{10}
All	High	9.20×10^5
Most	6.8%	1.06 (against)
Some	Low	—

UE Inferences (Superset)

Quantifier	"Yes" Rate	BF_{10}
All	Low	—
Most	2.3%	1.06 (against)
Some	High	3.50×10^8

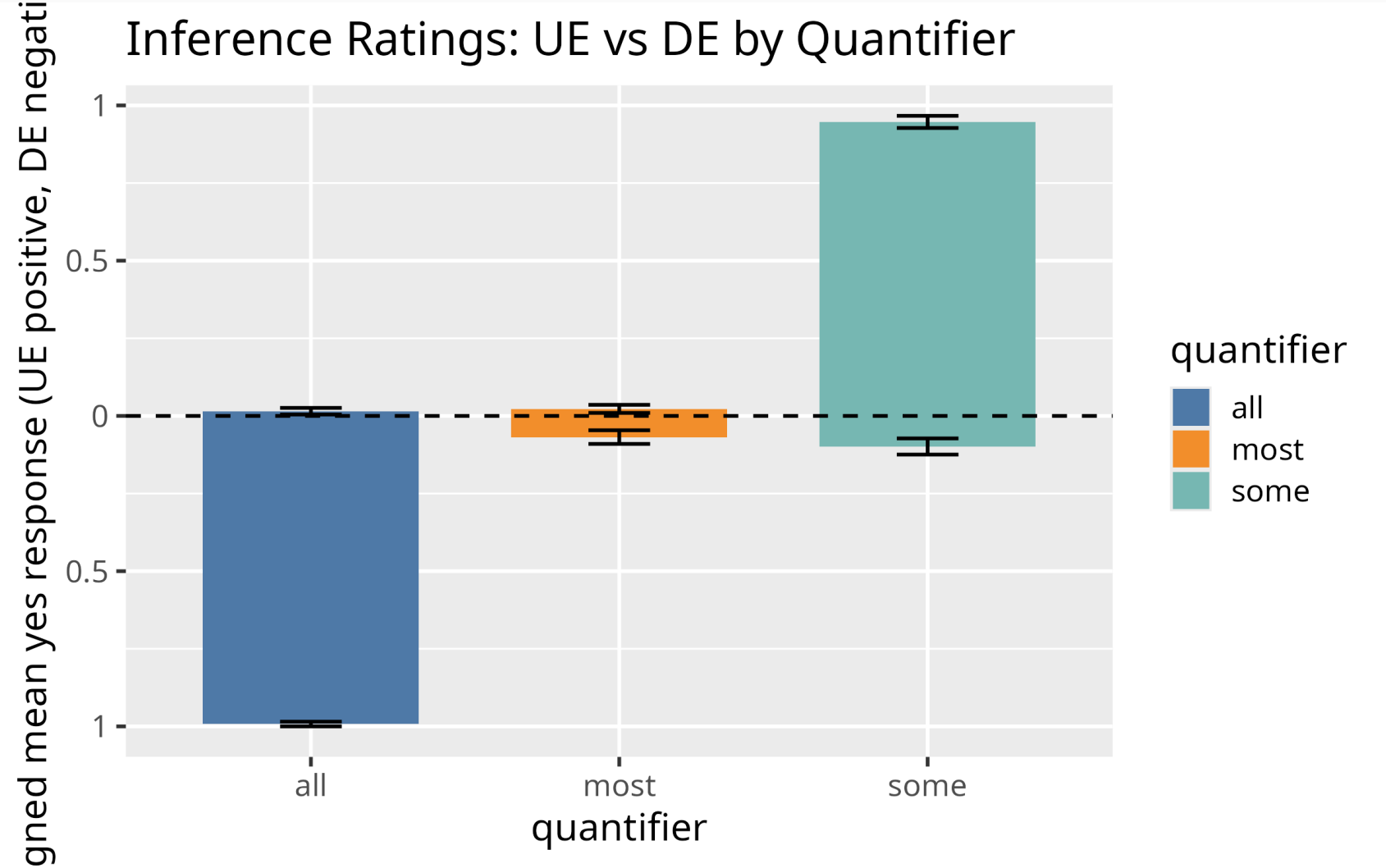


Figure 12: Inference rates by quantifier and direction

Critical: *Most* is Genuinely Non-Monotonic

- Both DE and UE rejected at near-floor levels
- BF_{01} = 1.06 for direction effect → **no credible preference**

The Dissociation

Acceptability vs. Inference

The Critical Pattern

- *Most* licenses NPIs (μ = 3.95) **despite** showing no monotonic inference pattern
- DE: 6.8% “yes” | UE: 2.3% “yes”
- This dissociation is **stronger evidence** against DI–NPI link than DL environments

Why Stronger?

- DL environments involve monotonic operators with complex scope
- *Většina* is **empirically non-monotonic** in the same participants

Theoretical Implications

Against Ladusaw (1979)

- Downward entailment is **not synchronically necessary** for NPI licensing
- NPIs appear in genuinely non-monotonic contexts

For Barker (2018)

- NPIs signal **narrow scope** with respect to their licensor
- Licensed when wide scope does not entail narrow scope

Formal Account

- $\exists d\,[\textit{většina}_x(F(x,d) \rightarrow P(x))]$?
- $\textit{většina}_x((\exists d\, F(x,d)) \rightarrow P(x))$?
- NPI takes narrow scope within restrictor (*sebemenší vírou*)

Diachronic vs. Synchronic

Diachronic (Herburger 2023)

- NPIs may have **grammaticalized in DE environments**
- Historical link to downward inference

Synchronic (Our Data)

- Downward inference **not necessary** for licensing
- Only **narrow scope** is required

The Acceptability Gradient

- Mirrors monotonicity landscape:
  - Maximal DE → highest acceptability
  - Genuine NM → intermediate acceptability
  - Maximal UE → lowest acceptability
- Compatible with diachronic grammaticalization + synchronic scope marking

Comparison: DL vs. NM Environments

Feature	Double Licensing (2025)	*Většina* (2026)
Operators	*doubt*, *negation* (monotonic)	*most* (non-monotonic)
Licensing	Domain-sensitive (BF > 10^{15})	Intermediate (μ = 3.95)
Inference	Uniformly UE (no domain effect)	No DE/UE pattern (BF_{01} = 1.06)
Dissociation	Licensing ≠ Inference	Licensing ≠ Monotonicity
Evidence Strength	Strong	**Stronger** (genuinely NM)

Conclusion

- Both studies support **scope over monotonicity**
- *Většina* provides the **critical test case**

Summary: Appendix Takeaways

1. **NPIs license in non-monotonic contexts** (*většina* μ = 3.95)
2. **No monotonic inference pattern** for *většina* (DE: 6.8%, UE: 2.3%)
3. **Dissociation confirmed**: Licensing ≠ Downward Inference
4. **Supports Barker (2018)**: NPIs mark narrow scope, not DE
5. **Stronger evidence** than DL environments (genuinely non-monotonic quantifier)
6. **Diachronic note**: DE may be historical, not synchronic requirement

Bottom Line

"Downward inference is not synchronically necessary for NPI licensing—only narrow scope is required."

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